NMC 2014 PRIMARY LEVEL PAPER 1 QUESTIONS

SECTION A

Qn. 1. How many days are in n weeks and m days?

Qn. 2. If $\frac{144}{0.144} = \frac{14.4}{x}$ then $x =$_____

Qn. 3. Arrange in descending order: $\frac{3}{7},\ \frac{5}{6},\ \frac{1}{3},\ \frac{6}{7},\ \frac{2}{5},\ \frac{4}{5}$

Qn. 4. The difference between two numbers is 1365. On dividing the larger number by the smaller number, we get 6 as quotient and 15 as the remainder. What is the smaller number?

Qn. 5. Which of the following is not a prime number and why: 31, 61, 71, 91?

Qn. 6. Given that:
S1:Joel is younger than Cathy
S2:Mark was born after Joel but before Cathy
S3:Cathy is younger than Mark
If statements S1 and S2 are true, then statement S3 is (*circle the correct answer*)
A) TRUE *B*) FALSE *C*) I DONT KNOW

Qn. 7. A circle of radius r and a square of side l have the same area of 154cm^2. The ratio $r^2 : l^2$ is

Qn. 8. An electrician places a ladder against a wall at a point 6m from the flat ground. The flat ground and the wall meet at right angles. If the ladder is 8m from the wall, its length is

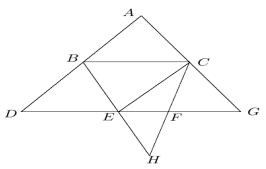
Qn. 9. A bus moves non-stop from town A to town B at 54km/h but if it makes bus stops along the way, its speed is 45km/h. How many more minutes does the bus require to travel from A to B?

Qn. 10. Mrs. Musoke presented a cheque to Stanbic Bank to withdraw five million five thousand five shillings only. She correctly wrote the amount in words on the cheque but in figures she wrote Shs. 5,050,005. The teller did not bother much with the words and went ahead and paid her Shs. 5,050,005. Did the bank make any loss? if so, how much loss was made?

SECTION B

Qn. 11. Examinations are taking place in two rooms A and B. If 10 candidates are moved from A to B, then the number of candidates in each room is the same. However, if 20 candidates are moved from B to A, then the number of candidates in A is double the number of candidates in B. How many candidates are in room A?

Qn. 12.



(a) Write down all the triangles in the figure.

(b) How many are they?

Qn. 13. Muwanga does 80% of work in 20 days. He then calls in Namuli and they together finish the remaining work in 3 days. How long would Namuli alone take to do the whole work?

Qn. 14. Three premier league footballers are paid per week in pounds. The average weekly income of Paul and Qually is 5050. The average weekly income of Qually and Robert is 6250 and the average weekly income of Paul and Rose is 5200.

(a) What is the total weekly income for all the players?

(b) What is the total weekly income of each of the players?

Qn. 15. Three girls (Joan, Jose, Julian) enter a cycling competition around a circular path. They take the path in turns. One is free to repair their bicycle and continue with the race provided one does not spend more than 10 minutes repairing the bicycle. Joan rides at 20km/h and takes 24 minutes.

(a) What is the length of the circular path?

(b) If Jose took 32 minutes, what speed did she ride?

(c) What is the diameter of the path?

(d) Julian starts with a speed of 25km/h but after 20 minutes, gets a problem that takes her 3 minutes to fix. At what speed must she ride in order to win the competition?

NMC 2014 PRIMARY LEVEL PAPER 1 SOLUTIONS

SECTION A : 5 marks each

1. There are 7 days in one week and hence $7n$ days in n weeks. Therefore there are $7n + m$ days in n weeks and m days.

2.

$$\frac{144}{0.144} = \frac{14.4}{x} \Longrightarrow 144x = 14.4 \times 0.144$$

$$x = \frac{14.4 \times 10 \times 0.144 \times 1000}{144 \times 10 \times 1000} = \frac{144 \times 144}{144 \times 10000} = \frac{144}{10000} = 0.0144.$$

3. The Lowest Common Multiple of $7, 6, 3$ and 5 is 210. We thus have the following.
   $\frac{3}{7} \times 210 = 90$, $\frac{5}{6} \times 210 = 175$, $\frac{1}{3} \times 210 = 70$, $\frac{6}{7} \times 210 = 180$, $\frac{2}{5} \times 210 = 84$, $\frac{4}{5} \times 210 = 168$. The arrangement of these fractions in descending order is $\frac{6}{7}, \frac{5}{6}, \frac{4}{5}, \frac{3}{7}, \frac{2}{5}, \frac{1}{3}$.

4. Let the numbers be x and y where $y < x$. Then $x - y = 1365$ and $x = 6y + 15$. Hence $6y + 15 - y = 1365 \Longrightarrow 5x = 1350 \Longrightarrow x = 270$. Therefore the smallest number is 270.

5. The factors of 31 are 1 and 31. Factors of 61 are 1 and 61. Factors of 71 are 1 and 71. Factors of 91 are $1, 7, 13$ and 91. Therefore 91 is not a prime number since it has more than two factors.

6. False. **Reason.** If S1 is true that Joel is younger than Cathy and S2 is true that Mark was born after Joel but before Cathy. Then Joel is the youngest of the three, followed by Mark while Cathy is the eldest. Hence it is false to say that Cathy is younger than Mark.

7. $\pi r^2 = 154$ and $l^2 = 154$. Hence $r^2 : l^2 = \frac{154}{\pi} : 154 = \frac{1}{\pi} : 1 = \frac{7}{22} : 1 = 7 : 22$.

8. Let the length of the ladder be l. By Pythagoras Theorem, $l^2 = 6^2 + 8^2 = 36 + 64 = 100$. Hence $l = \sqrt{100} = 10$.

9. Due to bus stops, it covers $54 - 45 = 9$ km less. So the time taken to cover 9 km is $\frac{9}{54} \times 60 = 10$ minutes.

10. Yes, the bank made a loss of Shs $5,050,005-$ Shs $5,005,005 =$ Shs $45,000$.

11. Let x be the original number of candidates in room A while y be the number of candidates in room B. Then $x - 10 = y + 10 \Longrightarrow x = y + 20$. Also $x + 20 = 2(y - 20)$. Solving these two equations together, we have $y + 20 + 20 = 2y - 40 \Longrightarrow 2y - y = 20 + 20 + 40 \Longrightarrow y = 80$. Therefore $x = y + 20 = 80 + 20 = 100$.

12. (a) The triangles are $ABC, BDE, BEC, CEF, CFG, FEH, CEH, CEG, BCH$ and ADG.

    (b) There are 10 triangles.

13. In 20 days, Muwanga does 80% of the work. Therefore he does $\frac{80}{20} = 4\%$ of the work in 1 day. Since the remaining 20% of the work is done jointly by Muwanga and Namuli in 3 days, it means they both do $\frac{20}{3}\%$ of the work in one day. This means that Namuli does $\frac{20}{3} - 4 = \frac{20-12}{3} = \frac{8}{3}\%$ of the work in one day. Hence 100% of the work will be done by Namuli in $\frac{3}{8} \times 100 = 37.5$ days.

14. (a) Total weekly income for all the players is $5050 + 6250 + 5200 = 16500$ pounds.

    (b) Let P, Q, R represent the total weekly income of Paul, Qually and Robert respectively. Then

$$\frac{P+Q}{2} = 5050 \Longrightarrow P + Q = 10100. \tag{1}$$

$$\frac{Q+R}{2} = 6250 \Longrightarrow Q + R = 12500. \tag{2}$$

$$\frac{P+R}{2} = 5200 \Longrightarrow P + R = 10400. \tag{3}$$

    Subtracting Equation (1) from Equation (2) gives $R - P = 2400$. This equation added to Equation (3) gives $2R = 12800 \Longrightarrow R = 6400$. Hence $P = 10400 - 6400 = 4000$ and $Q = 12500 - 6400 = 6100$.

15. (a) Distance = speed $\times$ time $= 20 \times \frac{24}{60} = \frac{480}{60} = 80$ km.

    (b) Speed= $\frac{\text{Distance}}{\text{time}} = 8 \div \frac{32}{60} = 8 \times \frac{60}{32} = \frac{480}{32} = 15$ km/h.

    (c) Circumference $= \pi d \Longrightarrow d = \frac{\text{circumference}}{\pi} = \frac{8}{\frac{22}{7}} = \frac{8\times 7}{22} = 2\frac{6}{11}$km.

    (d) In order for Julian to win the race, she must take less than 24 minutes to cycle around the circular path. Julian got a problem after covering a distance of $= 25 \times \frac{20}{60} = \frac{500}{60} = 8\frac{1}{3}$ km. Therefore this is not possible as Julian would have already completed the journey before the bicycle broke down.